Atharva Jadhav

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Summary

Machine Learning Engineer with hands-on experience spanning NLP, LLMs, computer vision, distributed training, and production MLOps. Skilled in building and deploying end-to-end ML systems using PyTorch, Hugging Face Transformers, MLflow, Docker, and FastAPI.

Skills

- **AI/ML:** End-to-end ML pipelines, predictive modeling, deep learning, feature engineering, statistical analysis, model evaluation and tuning, distributed PyTorch training (DDP, FSDP, DeepSpeed-ZeRO), medical AI workflows
- **NLP & Computer Vision:** Hugging Face Transformers, DistilBERT, InLegalBERT, BERT-based fine-tuning, text classification, NER, OCR, document parsing, CNNs, transfer learning, image classification, Grad-CAM explainability.
- **Tools & Frameworks:** Python, PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, Pandas, NumPy, SQL, Git, Linux.
- **MLOps & Deploymen:** MLflow, Docker, Docker Compose, FastAPI, Flask, REST APIs, WebSockets, model versioning, experiment tracking, GitHub Actions, CI/CD, AWS.

Projects

Multilanguage Image Captioning, Clemson SC Link GitHub *March 2026 - Present*
- Built a CLIP-Prototypical few-shot classifier achieving 92.5% accuracy from 3 examples/class; deployed via FastAPI/Streamlit.
- Added embedding-drift checks flagging beer mug - coffee mug ambiguity; tracked runs with MLflow/Evidently.

Distributed LLM Training Pipeline, Clemson SC Link GitHub *Oct 2025 - Jan 2026*
- Engineered modular training system with DDP, FSDP, DeepSpeed-ZeRO achieving 4× GPU scaling efficiency gains.
- Optimized multi-GPU throughput and convergence through systematic model configuration experimentation.

Emotion Classifier using DistilBERT, Clemson SC Link GitHub *Aug 2025 - Oct2025*
- Fine-tuned DistilBERT with Hugging Face and PyTorch for multi-class emotion classification.
- Built a reusable inference pipeline for real-time text emotion prediction and deployment.

Multimodal Legal Document Analyzer, Clemson SC Link GitHub *Jun 2025 - Aug 2025*
- Created production ML pipeline with OCR, NER, transformers, scheduled training, MLflow versioning, drift checks.
- Deployed secure containerized inference API with MongoDB integration and CI/CD for compliant document processing.

Heart Disease Prediction Using Machine Learning, Pune India Link GitHub *Jun 2021 - May 2022*
- Engineered end-to-end ML pipeline for cardiovascular risk prediction with hyperparameter tuning and validation.
- Deployed Flask-based inference API with MLflow experiment tracking ensuring reproducible training workflows.

Work Experience

Machine Learning Researcher – NanoBio Mechanics and Manufacturing Lab *Oct 2025 - Present*
Clemson SC
- Developed ML models for PFAS filtration prediction using SEM-derived morphology features and nanofiber data.
- Built feature extraction pipelines for data-driven nanofiber materials optimization across filtration and packaging research.
- Prototyping deep learning and FT-Transformer models for chitosan nanofiber food packaging property prediction.

Education

- **M.S in Computer Science**, Clemson University, SC, USA *(May 2025)* – GPA: 3.73/4.0
- **B.E in Computer Engineering**, SPPU, India *(June 2022)* – GPA: 3.46/4.0

Certifications

- TCS MasterCraft DevPlus - TCS (Development and engineering fundamentals)
- Accenture Developer Program - Forage (software engineering foundations)
- Agile Project Management - Udemy (Scrum, sprint planning, backlog execution)

Leadership and Involvement

- Research collaboration with cross-functional teams including materials scientists and domain experts
- Peer code review participation ensuring quality standards and knowledge sharing across projects